

LESSON 6-2 • SESSION 1 • PRACTICE • FORM A

Dividing a Fraction by a Fraction

Learning target – I can divide a fraction by another fraction using the reciprocal. · 6.NS.A.1

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**:
 $a/b \div c/d = a/b \times d/c$.
- Keep the first fraction, change $\div$ to $\times$, flip **only the second** fraction.
- Check:** quotient $\times$ divisor = the amount you started with.

WORKED EXAMPLE

A pitcher has $2/3$ liter of juice. Each cup holds $1/6$ liter. How many cups can be filled?

Step 1 Write the division: $2/3 \div 1/6$.

Step 2 Flip the divisor and multiply: $2/3 \times 6/1 = 12/3$.

Step 3 Simplify: $12/3 = 4$. Check: $4 \times 1/6 = 4/6 = 2/3$.

Answer 4 cups

WORDS reciprocal (*recíproco*) · dividend (*dividendo*) · divisor (*divisor*) · quotient (*cociente*)

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. $5/6 \div 1/3 =$ _____

b. $2/3 \div 4/9 =$ _____

c. $8/9 \div 2/9 =$ _____

d. $3/5 \div 1/10 =$ _____

e. $1/3 \div 5/6 =$ _____

f. $5/8 \div 3/4 =$ _____

2. **Soup and Nuts.** A shop has $3/4$ gallon of soup and $4/5$ pound of nuts. The bar is 1 gallon; the shaded part is the soup.



a. Each bowl holds $1/12$ gallon. How many bowls can the shop fill? _____ bowls

b. Each bag holds $2/5$ pound. How many bags of nuts? _____ bags

3. **Sanding Strips.** Lia has $11/12$ meter of sandpaper. She tears it into strips that are each $1/12$ meter wide. How many strips?

Hint Count the twelfths in $11/12$.

Show your work.

Answer _____ strips

4. **Honey Jars.** A tub holds $7/8$ pound of honey. Each gift jar holds $1/16$ pound. How many jars can be filled?

Hint $1/8$ pound holds two $1/16$ pounds.

Show your work.

Answer _____ jars

5. **Wire Pieces.** A $5/12$ -meter wire is cut into pieces that are each $5/12$ meter long. Nadia says there are $25/144$ pieces. Justify or correct her answer.

Hint Did Nadia flip the divisor? How many $5/12$ s fit in $5/12$?

Show your work.

Answer _____ piece

6. **Wood Stain.** A can holds $15/16$ liter of stain. Grant uses $3/16$ liter on a chair. He pours the rest into jars that each hold $1/16$ liter.

Hint Subtract first, then divide by $1/16$.

- a. How much stain is left? _____ L
b. How many jars does it fill? _____ jars

Show your work.

7. **Craft Supplies.** A roll holds $9/10$ meter of tape. A large project uses $3/10$ meter. A small project uses $1/10$ meter.

- a. How many large projects can one roll supply? _____ projects
b. How many small projects? _____ projects
c. Explain why the second answer is 3 times the first. _____

8. **Name Tags.** A roll holds $5/6$ yard of trim. Each name tag uses $1/18$ yard of trim.

Hint A tiny divisor means many groups fit inside.

- a. How many name tags can the roll cover? _____ tags
b. Explain why the quotient is so large. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Jamal's Work.** Jamal solved $3/5 \div 1/2$. He flipped both fractions and wrote $5/3 \times 2/1 = 10/3$. Explain his mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $3/4 \div 1/4$.

A $3/16$ B 3 C $1/3$ D $4/3$

11. To divide by a fraction, you multiply by its –

A numerator B denominator C reciprocal D opposite

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

7/128 Z	9 O	25/144 Q	15 R	1 V	3/4 I	12 S	5/108 X	14 I	11 D
#3		#4	#5	#6a	#6b	#7b	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____